

M708 HW 01

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Exercise 11

Let's assume the typical inverse function theorem from multivariable calculus and work from there.

Without loss of generality, we can assume $p = 0$, $F(p) = 0$, and U, V are small neighborhoods of the origins in $\mathbb{C}^n, \mathbb{C}^m$. Our assumption that $F_*(0)$ has rank k implies that we can permute the coordinates such that the matrix with entries $\partial F_i / \partial z_j$ ($1 \leq i, j \leq k$) is nonsingular. In these coordinates, we will regard $\mathbb{C}^n$ as $\mathbb{C}^k \times \mathbb{C}^{n-k}$ with coordinates (z, w) and $\mathbb{C}^m$ as $\mathbb{C}^k \times \mathbb{C}^{m-k}$. With this in mind, write $F = (F^1, F^2): \mathbb{C}^n \rightarrow \mathbb{C}^k \times \mathbb{C}^{m-k}$.

Define a map $\tilde{F}: \mathbb{C}^n \rightarrow \mathbb{C}^n = \mathbb{C}^k \times \mathbb{C}^{n-k}$ by $\tilde{F}(z) = (F^1(z, w), w)$. Clearly, by construction, this map's differential $\tilde{F}_*(0)$ has rank n at 0. Note that forgetting the complex structure, we have a smooth map $\tilde{F}$ between open subsets of $\mathbb{R}^{2n}$, whose differential at the origin has real rank $2n$. Hence, by the inverse function theorem, we can shrink U around 0 to some connected open neighborhood U' such that $\tilde{F}|_{U'}$ is a bijection onto some open set A with smooth inverse $\varphi: A \rightarrow U'$. We should verify now that φ is, in fact, holomorphic:

$$d\text{Id}_u = d(\tilde{F} \circ \varphi)_u = d\tilde{F}_{\varphi(u)}[d\varphi_u]$$

Since $d\tilde{F}_{\varphi(u)}$ is an invertible $\mathbb{C}$ -linear map (since $\tilde{F}$ is clearly holomorphic by construction), and so is the identity $d\text{Id}_u$, we conclude that $d\varphi_u$ is $\mathbb{C}$ -linear too, for all $u \in A$, so φ is (bi)holomorphic (and so is $\tilde{F}$).

By construction of $\tilde{F}, \varphi$, we have that

$$(z, w) = (\tilde{F} \circ \varphi)(z, w) = (F^1(\varphi(z, w)), (\varphi_{k+1}(z, w), \dots, \varphi_n(z, w)))$$

from which we conclude in particular that $F^1(\varphi(z, w)) = z$, that is, $F^1 \circ \varphi: \mathbb{C}^n \rightarrow \mathbb{C}^k$ is projection onto the first k coordinates.

It remains to deal with the rest of the coordinates. Note that since $d(F \circ \varphi)_{(z, w)}$ is already of rank k on just the first k coordinates with the upper-left matrix being the identity and to its right all zeros, the differential $d(F^2 \circ \varphi)_{(z, w)}$ must be zero on the w part, meaning $H(z) = F^2(\varphi(z, w))$ is independent of w . Thus we define another biholomorphism $\psi: V' \rightarrow B$ where V' is some neighborhood of the image of U' under F and B is its image under this mapping, given by $(z, w) \mapsto (z, w - H(z))$. This is clearly holomorphic with holomorphic inverse, so what we've said is well-defined.

Now we check that

$$\psi(F(\varphi(z, w))) = \psi(z, H(z)) = (z, H(z) - H(z)) = 0$$

Therefore, we have found biholomorphic $\varphi: A \rightarrow U', \psi: V' \rightarrow B$ such that

$$(\psi \circ F \circ \varphi)(z_1, \dots, z_n) = (z_1, \dots, z_k, 0, \dots, 0)$$

as a function from $A \subseteq \mathbb{C}^n$ to $B \subseteq \mathbb{C}^m$. This solves our original problem modulo affine maps for translation and choice of coordinates, which are definitely holomorphic.

Exercise 12

First, we claim that $\det: \mathbb{C}^{n \times n} \rightarrow \mathbb{C}$ is holomorphic of constant rank 1 in a neighborhood of any point $X \in SL_n(\mathbb{C})$. It's holomorphic because it's a polynomial, and since its rank is at most 1, we just have to check that its differential $d \det_X$ doesn't vanish at X . We have the well-known Jacobi formula for the derivative:

$$\frac{\partial \det}{\partial X_{ij}}(X) = (-1)^{i+j} M_{ij}$$

where M_{ij} is the (i, j) th minor of X . We can also cofactor-expand $\det X$:

$$1 = \det(X) = \sum_{i=1}^n (-1)^i M_{i,1}$$

where the first equation is from $X \in SL_n(\mathbb{C})$. Thus not all the $M_{i,1}$ terms are zero, so some $M_{i,j} \neq 0$, so $d \det_X \neq 0$. Hence, the claim holds.

Now we can apply the previous exercise: around each point $X \in SL_n(\mathbb{C})$, we obtain neighborhoods $X \in U \subseteq \mathbb{C}^{n \times n}$ and $1 \in V \subseteq \mathbb{C}$, and biholomorphic maps $\varphi: U \rightarrow A$, $\psi: V \rightarrow B$ such that $\psi \circ \det \circ \varphi^{-1}: A \rightarrow B$ projects onto one coordinate. The preimage of $\psi(1)$ under this map is the intersection of A with some affine linear subspace of $\mathbb{C}^{n \times n}$, which we can identify with an open subset U' of $\mathbb{C}^{n^2-1}$. By construction, φ restricts to a biholomorphic map taking U' to $U \cap SL_n(\mathbb{C})$, providing us a local chart around X . Having done this at each $X \in SL_n(\mathbb{C})$, it's clear that the transition maps are biholomorphic by construction. Thus, $SL_n(\mathbb{C})$ is a complex manifold.

Multiplication is holomorphic since in charts, it's given by restricting a composition of holomorphic functions (the biholomorphic maps from above, with ordinary matrix multiplication in the middle) to a subset of the form of a product of U 's as above. Inversion is holomorphic for the same reason, keeping in mind that ordinary matrix inversion is holomorphic away from the zero matrix due to the adjugate formula.

Exercise 13

We must define chart maps $\varphi_I: U_I \rightarrow \mathbb{C}^{k(n-k)}$ (I claim this is the right dimension and the charts can be the whole affine space). Define φ as follows: let $W \subseteq \mathbb{C}^n$ be a subspace which can be written as the row space of a $k \times n$ matrix X with nonvanishing minor indexed by the columns in I . Let $Y \in GL_k(\mathbb{C})$ be the $k \times k$ matrix given by the I th columns of X . Then $Y^{-1}X$ has the identity matrix in the I th columns, and we define $\varphi(X)$ to be the $k \times (n-k)$ matrix obtained by deleting these columns. This is well-defined because $GL_k(\mathbb{C})$ acts freely and transitively on bases of a given k -dimensional subspace W , so there is a unique representative matrix with the identity in the columns indexed by I . The inverse is obvious (insert identity matrix at positions I), and it's clear that both maps are continuous, so it's a homeomorphism.

Now to check the transition functions: from U_I to U_J , we take a $k \times (n-k)$ matrix, insert some columns, multiply by the inverse of a matrix given by some of the values, then project down by removing some columns. This is clearly holomorphic: we know linear inclusions, matrix multiplication, matrix inversion, and linear projections are all holomorphic. Thus, $G(k, n)$ is a complex manifold.

To see that it is compact, we note that it's a quotient of $(S^{2n-1})^k$ (which is compact), defined by taking the span of a given k unit vectors in $\mathbb{C}^n$. This is clearly a continuous quotient map, in

particular surjective since we can normalize any basis of any k -dimensional subspace. Since a quotient of a compact space is compact, $G(k, n)$ is a compact complex manifold.